

# Mohamed Zakaria Elnemr

📍 Cairo, Egypt | 📞 +20 102 499 5208 | ✉ malnemr0@gmail.com  
🌐 linkedin.com/in/tiger0x01 | 🐙 github.com/Tiger0x01 | 📊 kaggle.com/tiger0x01

## SUMMARY

Final-year Computer Science student specializing in Machine Learning and Generative AI, with hands-on experience building RAG applications, LLM-powered systems, computer vision solutions, and end-to-end ML pipelines through Microsoft DEPI, internships, and practical AI projects.

## EXPERIENCE

**Microsoft Machine Learning Engineer Trainee** | *DEPI* Jun 2025 – Dec 2025

- **Engineered** and deployed end-to-end ML pipelines on Azure, applying MLOps practices for model deployment, experiment tracking, versioning, and monitoring using MLflow.
- **Developed** Deep Learning models for NLP and Computer Vision tasks, applying transfer learning and attention architectures to improve inference accuracy.
- **Built** Generative AI solutions and RAG-based applications utilizing advanced Prompt Engineering techniques tailored for diverse datasets.

**Generative AI Intern** | *Edrak For AI* May 2026 – Jul 2026

- **Completed** a comprehensive 126-hour Generative AI internship focused on modern AI concepts, LLM applications, and practical Generative AI workflows.
- **Built** end-to-end AI applications leveraging LLMs, prompt engineering, and retrieval-augmented generation (RAG) to solve practical business scenarios.

**Python Track Trainee** | *Information Technology Institute (ITI)* Jan 2025 – Feb 2025

- **Developed** scalable backend web applications using Django, integrating user authentication and CRUD functionality.
- **Designed** relational database architectures using PostgreSQL and integrated them with Django ORM.

## SELECTED PROJECTS

**Bayan AI** | *RAG, Qwen 2.5, FastAPI, React, TypeScript* GitHub

- **Built** an AI-powered cross-cultural communication assistant that rewrites professional messages according to the target culture's etiquette while preserving the original meaning.
- **Implemented** a RAG pipeline utilizing FAISS to retrieve relevant cultural knowledge, construct contextual prompts, and generate tailored responses through a FastAPI backend.

**NosoosBot** | *Python, Aiogram, Groq, Llama 3.3 70B* GitHub

- **Developed** an asynchronous Telegram bot for extracting YouTube captions, translating transcripts, and generating AI-powered multilingual summaries.
- **Implemented** transcript caching, rate limiting, document exports (PDF/DOCX), and Map-Reduce summarization techniques to process long-form content efficiently.

**BHAAV – House Price Prediction** | *Scikit-learn, TypeScript, React* GitHub

- **Developed** an AI-powered platform for predicting house prices across the Indian real-estate market using machine learning regression algorithms.
- **Architected** the end-to-end ML workflow, from data preprocessing and model training to evaluation and API integration into a React-based web application.

**Hand Gesture Recognition System** | *Python, OpenCV, MediaPipe* GitHub

- **Engineered** a real-time computer vision system using Python and MediaPipe to accurately track and classify sign language gestures.
- **Implemented** image-processing logic using OpenCV to track and record multiple hand landmarks simultaneously for real-time inference.

## EDUCATION

**Mansoura University** | *B.Sc. in Computer and Information Sciences* Sep 2023 – 2027 (Expected)

- **Final Year (4th Year)** | **Cumulative Grade: Very Good (83.0%)**

- **Relevant Coursework:** Artificial Intelligence, Algorithm Analysis & Design, Software Engineering, Network & Computer Security.

## TECHNICAL SKILLS

---

**Languages:** Python, SQL, C++ , PHP

**ML/DL:** Scikit-learn, PyTorch, TensorFlow

**GenAI/NLP:** RAG, LLMs, Hugging Face, FAISS, Prompt Engineering

**Computer Vision:** OpenCV, MediaPipe

**Backend:** FastAPI, Django

**Cloud & MLOps:** Azure AI, MLflow, Docker, Git, Linux

## CERTIFICATIONS

---

|                                                                |                    |
|----------------------------------------------------------------|--------------------|
| <b>Fundamentals of Deep Learning</b> — NVIDIA                  | <i>Aug 2026</i>    |
| <b>Generative AI (Beginner Level)</b> — NVIDIA DLI & ITI       | <i>Sep 2025</i>    |
| <b>Computer Vision</b> — Kaggle Learn                          | <i>Aug 2026</i>    |
| <b>Intro to Deep Learning</b> — Kaggle Learn                   | <i>Aug 2026</i>    |
| <b>HCIA AI Learning Course (40 Hours)</b> — Huawei ICT Academy | <i>Summer 2025</i> |